AI 101 for Mechanical Engineers · Module 12

Represent the information before choosing a network

Lesson 12.1

What you will learn

> Define the data representation before choosing a network, relate feature vectors to MLPs and local spatial patterns to CNNs, and design a split for new-component inspection.

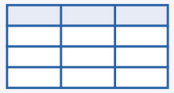
Lesson overview

Module 12 · Lesson 1

Represent the information before choosing a network

AI 101 for Mechanical Engineers

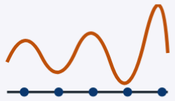
1 Describe the data first



Tabular data
Named feature values for each example.



Images
Numerical pixels and channels.



Signals
Values in time order, with timestamps, sampling information and units.

Record before training

1. Feature definitions and units
2. Sampling or image-acquisition conditions
3. Target and inputs available at decision time

2 Match structure to architecture

Multilayer perceptron (MLP)

Feature vector → Fully connected layers → Output

Convolutional neural network (CNN)

Pixels and channels → Shared local filters → Feature map

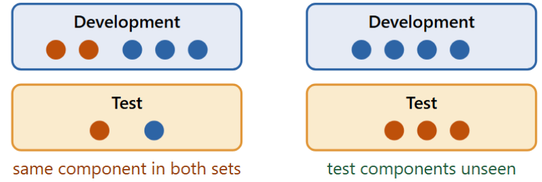
The same learned filter is applied across local regions, helping detect patterns such as edges or texture.

A filter response is a number, not a confirmed defect. Inspection labels and suitable validation are still required.

Architecture choice is a starting point, not a performance guarantee.

3 Test inspection on new components

Several views of the same component share surface patterns, lighting and acquisition settings.



For new hardware, hold out whole components or appropriate batches.

Check annotation quality and any changes in cameras or lighting.

Key takeaway **Preserve the relevant data structure, then choose the network and the evaluation split for the engineering question.**

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Three data forms, three representations

Data	Represented as
Tabular	Named feature values for each example
Image	Pixels and channels
Signal	Measured values with timestamps, sampling, units

> Converting data to an array does not mean the model understands its engineering meaning.

Document the representation before training

1. Feature definitions and units
2. Sampling or image-acquisition conditions
3. The target
4. Information available at the intended decision time
5. Preserve the structure relevant to the engineering question

Architecture follows data structure and required output

MLP

- Processes a feature vector
- Fully connected layers

CNN

- Shared filters over local regions
- Can represent local spatial patterns in images, some signals

> A starting design choice, not a guarantee of performance.

A filter response is not a defect finding



- The same filter is reused across locations [62]
- This can help detect edges or texture
- Not proof of a crack: labels and validation still needed

Match the split to the inspection question

Random image split

- Same component in training and testing
- Can overestimate performance [58]

Component holdout

- Whole components or batches held out
- Test components unseen during development

Model may recognise the component. Also consider annotation quality, lighting and camera changes.

Model output is a candidate finding

- Camera image: flag a suspected weld crack
- Thermogram: identify an unusual hot region for inspection
- Not a confirmed defect: conditions, labels, procedure need validation

CHECK YOUR UNDERSTANDING

Why can random image splitting overstate performance on new components?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why can random image splitting overstate performance on new components?

> Images of one component can share surface patterns and lighting across splits, so the model may recognise the component rather than generalise. A component-level holdout better represents the new-component question.

Key takeaways

- Tabular: feature vectors; images: pixels and channels; signals: timed values
- MLP for feature vectors; CNN for relevant local spatial patterns
- A filter response alone does not prove a physical defect
- Hold out whole components for new-component inspection

References

[58] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” Patterns, vol. 4, no. 9, Art. no. 100804, 2023. doi: 10.1016/j.patter.2023.100804

[62] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” Proc. IEEE, vol. 86, no. 11, pp. 2278–2324, 1998. doi: 10.1109/5.726791

Numbers match the course reference list. The written lesson is the authoritative text.